Satiyabooshan Murugaboopathy

Personal Information

Dresden, 6th May 1998
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Research Profile

Machine learning researcher with hands-on **automatic speech recognition** experience, including production **Whisper-based German ASR** pipelines and experimentation with **self-supervised speech representations**. Core strength in **uncertainty-aware, low-resource modeling** and **systematic benchmarking** - precisely the conditions of historical dialect archives : regional variation, spontaneous speech, and scarce annotated data. Motivated by speech technology that serves both **cultural-heritage preservation** (archival dialect corpora) and **clinical research** (speech as a non-invasive digital biomarker).

Research Interests

- **Low-Resource & Uncertainty-Aware Speech Technology :** Robust ASR and transcription for low-resource languages ; self-supervised speech representations ; confidence estimation and error analysis
- **Data-Efficient Machine Learning :** Transfer and few-shot learning for data-scarce settings ; systematic benchmarking and **open benchmark dataset** construction
- **Sequence & Multimodal Modeling :** Temporal modeling of streaming audio and sensor data ; integrating heterogeneous modalities (text, vision, audio, geospatial) into predictive models

Research Experience

Linköping University - Institute for Analytical Sociology (IAS)
Research Engineer *2026-Present*

- Developing robust **uncertainty-aware ML pipelines and prediction models** using advanced transfer learning techniques for complex, sparse-data settings
- Engineering **multimodal computational approaches** to bridge machine learning methodology with domain-specific scientific and industry questions

Voice Agent & On-Device ASR Venture
Co-Founder & AI Engineer *2021-Present*

- Built production **Whisper-based ASR pipelines** with streaming audio chunking, low-latency inference, and privacy-preserving on-premise execution
- Experimented with **self-supervised speech representations** (wav2vec 2.0, HuBERT) and **CTC-based decoding** for transcription and intent detection in low-resource settings
- Engineered **streaming voice interaction logic** (turn-taking, interruptions, dialogue state management), combining ASR and TTS into end-to-end voice agents with fine-grained control over tool use and workflow execution
- Optimized **whisper.cpp for Apple Metal** for on-device inference, building on an iOS transcription app focused on local, low-latency speech processing

Fraunhofer Society - IMW and ISI
AI Research Assistant *2024-2026*

- Building scalable **NLP pipelines** for computational text analysis on scientific and social discourse using advanced methods
- Implementing **automated systematic review** workflows combining retrieval and generation on **large, noisy, heterogeneous corpora**
- Leading **predictive trend analysis** projects using weak supervision methods to identify emerging patterns in noisy data

Porsche AG - Emerging Technologies Research
Interdisciplinary AI Research Assistant *2021-2024*

- Collaborated with **interdisciplinary teams** including psychologists, engineers, and UX designers to develop production-ready, human-centered AI systems
- Engineered scalable transfer learning pipelines for **behavioral analysis** using **sequence and temporal modeling** on streaming multimodal sensor data in data-scarce environments
- Developed **recommendation systems** using domain-specific language models to understand user preferences and social influence patterns

University Leipzig - Institute of Computer Science
Research Assistant - Logic and Computational Methods *2022-2023*

- Authored comprehensive **lecture materials on Advanced Logic**, demonstrating ability to communicate complex methodological and engineering concepts
- Contributed to pedagogical development in computational reasoning methods applicable to applied research

Fraunhofer Society - IMW
Research Assistant - Data Integration for Social Analysis *2020-2022*

- Engineered **automated database fusion pipelines** using NLP techniques for integrating heterogeneous data sources
- Developed scalable methodologies for extracting **structured insights** from scattered, multi-source datasets

Volkswagen AG - Software Development Center
Applied AI Research Assistant *2019-2020*

- Developed and optimized machine learning models for **pattern recognition in low-resource, extremely small-sample settings**, demonstrating expertise in data-scarce environments
- Presented engineering and research findings to diverse stakeholder groups, building skills in **cross-functional communication**
- Showcased as a flagship project by the Software Development Center Dresden on multiple occasions and named **one of the five most innovative AI projects** within the Volkswagen Group

Education

- M.Sc. in Computer Science, University Leipzig *2020-2026*
  Focus : Machine Learning, Data Science and Computational Methods
  *Master thesis : **Systematic benchmarking** of multi-label text classification methods on heterogeneous data under industrial constraints, including classical NLP, deep learning, large language models and small language models.*
- B.Sc. in Computer Science, University Leipzig *2017-2020*
  Focus : Machine Learning, Data Science and Data Analytics
  Bachelor thesis : AI-assisted optical quality control using very limited image data (>100 images)
- Abitur, Dresden, Germany *2008-2016*
  Physics courses at TU Dresden during high school years

Peer-Reviewed Publications

- Motion Detection in Videos of Coherence Matrices in order to detect Consciousness States in CLIS-patients - an Approach
  V. S. Adama, A. Blankenburg, C. Ernst, R. Kummer, S. Murugaboopathy and M. Bogdan
  10th EUROSIM Congress 2019 - ISBN 978-3-901608-92-6, DOI 10.11128/arep.58

- Technical Design Space Analysis for Unobtrusive Driver Emotion Assessment Using Multi-Domain Context
  D. Bethge, L. Falconeri, T. Kosch, S. Murugaboopathy, U. v. Zadow, A. Schmidt and T. Grosse-Puppedahl
  ACM IMWUT 2023, DOI 10.1145/3569466

- CARSI II : A Context-Driven Intelligent User Interface
  M. Wiedner, S. Naveenachandran, P. Hallgarten, S. Murugaboopathy and E. Frazzoli
  ACM AutomotiveUI '24, DOI 10.1145/3641308.3685036
  Best Poster Award *and* ***Best Work In Progress Award***
- KI-gestützte Themenlandkarte der Arbeitsforschung : Ein Kompass für die Zukunft der Arbeit ?
  P. Achkar, S. Murugaboopathy, A. Kreuter, T. Gollub, M. Potthast, Y. Campbell
  GfA Frühjahrskongress 2025, https ://publica.fraunhofer.de/handle/publica/517042
- From Keyterms to Context : Exploring Topic Description Generation in Scientific Corpora
  P. Achkar, S. Murugaboopathy, A. Kreuter, T. Gollub, M. Potthast, Y. Campbell
  EMNLP 2025 Workshop NewSumm, DOI 10.18653/v1/2025.newsum-main.8
- Platonic Representations for Poverty Mapping : Unified Vision-Language Codes or Agent-Induced Novelty ?
  S. Murugaboopathy, C. T. Jerzak and A. Daoud
  International Conference on Social Computing 2026, DOI 10.48550/arXiv.2508.01109
- Agent Verifiers in Counterfactual Analysis : A Pilot Study Benchmarking Agentic Heterogeneous Treatment Effects
  S. Murugaboopathy, Entejar Alam, Nicolas Audinet de Pieuchon, A. Daoud and C. T. Jerzak
  Accepted at NeurIPS 2026 Workshop VerifyAgents
- LUCID : Statistically Grounded Self-Correction for LLM Reasoning Evaluation
  S. Murugaboopathy, C. T. Jerzak, Nicolas Audinet de Pieuchon and A. Daoud
  Under Review at EACL 2027

Patents

- Method and apparatus for robust identification of a vehicle occupant
  S. Murugaboopathy, D. Bethge and M. Kari
  US Patent US20230147024A1 *Filed 11th May 2023*
- Verfahren und System für eine robuste Identifizierung eines Fahrzeuginsassen
  S. Murugaboopathy, D. Bethge and M. Kari
  German Patent DE10 2021 129 103B3 *Filed 8th December 2022*
- Real-Time Gaze and Pose based Rear and Side Mirror Adjustment using Computer Vision
  S. Murugaboopathy, D. Bethge, P. Hallgarten and P. Wolters
  German Patent DE10 2022 106 797B3 *Filed 27th April 2023*
- Verfahren, System und Computerprogrammprodukt zum Ausgeben von Ausgabewerten zur Analyse und/oder Bewertung und/oder Prozesssteuerung einer Entität
  S. Murugaboopathy, D. Bethge and M. Kari
  German Patent DE10 2021 129 108A1 *Filed 11th May 2023*
- Vorrichtung und Verfahren zur Fahrerassistenz für ein Kraftfahrzeug, die Vorrichtung umfassendes Kraftfahrzeug
  S. Murugaboopathy, D. Bethge and C. Patsch
  German Patent DE10 2022 104 322A1 *Filed 24th August 2023*
- Verfahren zur Herstellung eines Videofilms einer Fahrt eines Kraftfahrzeugs
  S. Murugaboopathy, P. Hallgarten, T. Grosse-Puppendahl and M. Kari
  German Patent DE10 2024 118 715.8 *Filing in Process*

Technical Skills

- **Programming : Python, R, SQL**, Swift, Java, C, HTML, Assembler, REXX
- **Machine Learning : PyTorch**, TensorFlow, Hugging Face, Transfer Learning, Few-Shot Learning, Uncertainty-Aware Modeling, Natural Language Processing, Multi-label Classification
- **Advanced ML Systems :** Multimodal Learning, Sequence and Temporal Modeling, Network Analysis, Scalable Data Integration Pipelines
- **Evaluation :** Systematic benchmarking, error and confidence analysis, hallucination analysis
- **Systems & Infrastructure :** Linux, Containerization, CI/CD, Distributed Systems

Languages

- **Native : German**, Tamil, English (C1)
- **Proficient :** French (B2), Latin (Latinum)

Interdisciplinary Experience

- **Cross-cultural & Cross-disciplinary Perspective :** Born in Germany to a Tamil family, year in India (2007-2008) - at ease across cultural and disciplinary contexts
- **Team Sports Leadership :** Semi-professional volleyball player in the German national-level beach and Saxony federal-level indoor-volleyball squad ; captained multiple teams *2005-2016*

Satiyabooshan Murugaboopathy

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Statement of Research Interests and Relevant Experience

Doctoral Researcher Position 1: Dialect-aware Automatic Speech Recognition

Automatic Swabian Recognition (ASR): Processing and Evaluating Dialect Speech

University of Tübingen - Cluster of Excellence "Machine Learning: New Perspectives for Science"

September 30, 2026

I am applying for the doctoral position on dialect-aware automatic speech recognition within the Automatic Swabian Recognition project. I am a machine learning researcher with hands-on ASR engineering experience and a methodological focus on the three pillars of the advertised dissertation: low-resource and uncertainty-aware modeling, self-supervised speech representations, and the construction of a curated open benchmark dataset. As a native German speaker with a track record of interdisciplinary collaboration, I am attracted to a project in which archival dialect recordings, computational linguistics, and clinical research meet. Below I outline my research interests, the experience I would bring, and my perspective on the dissertation.

Research Interests

Uncertainty-aware ASR for dialect, spontaneous, and historical speech. State-of-the-art ASR degrades under the conditions of the Arno Ruoff Archive: regional variation, spontaneous speech, historical recording technology, and scarce annotated data. I am very interested in how self-supervised speech representations (wav2vec 2.0, HuBERT, WavLM) can be adapted to such shifted domains, and how uncertainty quantification can make their outputs trustworthy for curation. I want to build transcription pipelines whose confidence estimates are calibrated on dialect speech, so that uncertain segments can be routed to expert review.

Benchmarking as a research contribution. The project's central deliverable, being an open, curated benchmark uniting computational linguistics, machine learning, and cultural anthropology, very much matches my methodological trajectory. My M.Sc. thesis systematically benchmarked text-classification methods on heterogeneous data under industrial constraints, spanning classical NLP, deep learning, and language models. My recent methodological work is likewise about evaluation: statistically grounded self-correction for evaluating LLM reasoning (LUCID, under review), and a pilot benchmark for agentic heterogeneous treatment effects (NeurIPS 2026 Workshop VerifyAgents).

Speech as a digital biomarker. The clinical aspect of the project, to validate dialect-aware models on the TREND cohort to establish speech as a non-invasive marker of cognitive change, is a direction I find very exciting. My first publication analyzed clinical data to detect consciousness states in completely locked-in patients (EUROSIM 2019), and I am really motivated by the potential impact of this work.

Interdisciplinary machine learning. Across my positions I have worked with sociologists, psychologists, engineers, and UX researchers, and I value projects where ML methods serve domain questions. The prospect of working between the Department of Linguistics and the Hertie Institute for AI in Brain Health, is very interesting to me.

Relevant Experience

Production ASR systems. As co-founder of a voice-agent venture (2025–present) I build a Whisper-based ASR pipelines for speech in production: streaming audio chunking, low-latency inference, and privacy-preserving on-premise execution, combined with TTS and dialogue-state management into end-to-end voice agents. I have hands-on experience with self-supervised speech representations and CTC-based decoding for transcription and intent detection in low-resource settings. Earlier, I optimized whisper.cpp for Apple Metal shortly after its release, building on an iOS transcription app I co-created in 2021.

Low-resource and uncertainty-aware machine learning. As a research engineer at the Institute for Analytical Sociology (Linköping University), I develop uncertainty-aware prediction pipelines and transfer-learning workflows for complex, sparse, heterogeneous data - the same modeling regime the Ruoff material inhabits. Earlier positions deepened this: pattern recognition in extremely small-sample settings at Volkswagen, and sequence and temporal modeling on streaming multimodal sensor data at Porsche.

Working with noisy historical-scale corpora. At Fraunhofer IMW and ISI I built NLP pipelines and automated review workflows over large, noisy, heterogeneous corpora, and led predictive trend analysis using weak supervision - experience that maps directly onto curation and transcription at archive scale (1,600 interviews). Six patents in intelligent systems and production-grade engineering practice (containerized, reproducible pipelines) support the release and maintenance of an open benchmark.

Perspective on the Dissertation

Building on this foundation, my starting vision for the dissertation is: (i) characterize how and why self-supervised speech models transfer to 1950–1975 Upper German dialect recordings, treating domain shift and diachronic language change as research objects; (ii) develop calibrated, uncertainty-aware ASR and IPA transcription with human-in-the-loop curation of expert annotations; (iii) release the open parallel-tier benchmark with documented baselines and error analysis; and (iv) validate robustness on contemporary clinical speech from the TREND cohort. I am fully aware that I have not yet worked with IPA transcription or dialect-annotated corpora, and I see close supervision by Prof. Jäger and Prof. Ritter as an opportunity to grow into these areas. What I bring is the methodological and engineering toolset as well as genuine interst to learn more about unfamiliar methods and materials. I am available to begin at short notice.

References

1. **Prof. Dr. Connor Thomas Jerzak** - Assistant Professor, Department of Government, University of Texas at Austin; co-PI, AI & Global Development Lab. Research collaborator and co-author.
   connor.jerzak@austin.utexas.edu

2. **Dr. Yuri Cassio Campbell Borges** - Deputy Head, Data Science for Innovation Group, Fraunhofer Institute for Systems and Innovation Research ISI. Research collaborator and co-author during my Fraunhofer appointment (2024–2026).
   yuri.cassio.campbell.borges@isi.fraunhofer.de